Snehangshu Roy

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Projects

quant — Event-Sourced Trading Engine
Tech: Rust, Tokio, Parquet/Arrow, PostgreSQL, Docker, zstd

- Built a **12-crate, 65K-line** Rust workspace in which one strategy binary runs unchanged across historical Parquet replay and a live venue feed, so a backtest cannot describe a system that was never run.
- Ran a **14-day live paper session** against Binance with capture and engine fed from one ingress, then backtested the same strategy over the data recorded during that window: **paper P&L reproduced the backtest exactly** — same fill count, cash and realized P&L to the satoshi, on both symbols — across **97.9M** recorded frames with **zero** missing.
- Enforced risk as a **mandatory chokepoint** rather than a library the strategy calls: under test an oversized order never reaches the venue, and a runaway strategy stays capped at its order limit for any run length. **797** tests green; money as fixed-point `i64` with float arithmetic denied workspace-wide.

Low-Latency Limit Order Book — C++ (team of 2)
Tech: C++17, perf (hardware counters), Docker, open-addressing hash tables, bitmaps

- Benchmarked three designs on 1M-op workloads (`std::map` + hash map, lazy-deletion heap, our flat price array + bitmap + open-addressing table); ours won every op at **56.7 ns** add / **38.3 ns** cancel, **3.8**× faster overall.
- Explained the gap with `perf stat` / `perf record` counters rather than guesses: no allocation per op, ∼**5**× fewer L1 misses and IPC **1.41** vs **0.79**; every op checked against a brute-force reference book.

RedPen — Rubric-Anchored Marking & Annotation
Tech: TypeScript, Node.js, React, pdf.js, pdf-lib, Anthropic Claude / Gemini / Groq, Vitest

- Marks answer papers against a rubric and anchors every mark to a verbatim student quote, mapping quote → character span → page rectangle so feedback is boxed where the mistake is, not at a coordinate the model invented.
- Trusted the model's judgement but not its arithmetic: invariants clamp each mark to its rubric row and reconcile the total, throwing rather than returning a confidently wrong score.
- Routed **four** model backends (Claude, Gemini, Groq/OpenAI-compatible, deterministic offline) behind one interface with retry and labelled fallback; **102** offline tests against self-authored ground truth of **23** planted errors.

Nexo — Real-Time Messaging Platform
Tech: TypeScript, Next.js, Node.js, Express, Socket.io, Redis, Prisma, PostgreSQL, Docker

- Engineered WebSocket messaging backed by **Redis pub/sub**, so message fan-out survives horizontal scaling across server instances instead of being pinned to one process's socket map.
- Modelled presence, typing indicators and sent/delivered/seen receipts on Prisma/PostgreSQL, alongside rich media (voice notes with deterministic waveforms), edits, and scheduled and self-destructing messages.

Sift — Document Processing & Hybrid Search
Tech: Python, FastAPI, Next.js, PostgreSQL/pgvector, Celery, Redis, S3, fastembed, Tesseract

- Fused PostgreSQL full-text search with pgvector semantic search via **Reciprocal Rank Fusion**, over embeddings generated locally on ONNX with no paid API.
- Built async **Celery** pipelines for text extraction, OCR fallback, thumbnailing and embedding, with per-document status pushed live to the client over WebSockets.
- Designed multi-tenant org/RBAC with team invites, JWT auth, S3-compatible storage and Alembic migrations; the same code runs on local Docker and managed free tiers via env config.

Education

Scaler School of Technology — **2024 – Present**
Undergraduate Program in Computer Science — *Bengaluru, India*
- CGR: **8.31**

Birla Institute of Technology (BITS) — **2024 – Present**
B.Sc. (Hons.) in Computer Science — *Online, India*
- CGPA: **8.2**

Technical Skills

Languages	Rust, TypeScript, Python, Java, C++, SQL
Backend & Frameworks	Node.js, Express, FastAPI, Socket.io, Prisma, SQLAlchemy, Celery
Frontend	Next.js, React, Tailwind CSS, TanStack Query, Zustand
AI & LLM Tooling	Claude, Gemini, Groq, multi-provider routing & fallback, LLM-as-judge evals, RAG, vector search
Systems	Event-sourced design, low-latency C++, perf profiling, Redis pub/sub, WebSockets, async job pipelines
Databases	PostgreSQL, pgvector, Redis, SQLite, Parquet/Arrow
Cloud & DevOps	Docker, Kubernetes, Helm, Terraform, AWS (EKS, EC2, S3), Argo CD, GitHub Actions, Prometheus
Testing	Vitest, pytest, cargo test, Playwright, deterministic offline harnesses